

Software Development Life Cycle (SDLC) & Agile Methodology

Self-Study Guide

1. SDLC Fundamentals

Definition: SDLC is a structured process for planning, developing, testing, deploying, and maintaining software.

Phases: Requirements, Design, Development, Testing, Deployment, Maintenance.

Example: Banking application – gather customer requirements, design database, implement fund transfer, test transactions, deploy to production, maintain with security patches.

Mini Case Study: An e-commerce company discovers checkout failures after launch because testing was skipped. Which SDLC phase was weak?

2. SDLC Models

Waterfall, V-Model, Iterative, Spiral, Incremental, Agile.

Example: ATM firmware uses Waterfall due to stable requirements; Startup mobile app uses Agile due to changing customer needs.

Exercise: Match the project with the most suitable SDLC model and justify your choice.

3. Agile & Scrum

Agile values collaboration, customer feedback, and iterative delivery.

Roles: Product Owner, Scrum Master, Developers.

Artifacts: Product Backlog, Sprint Backlog, Increment.

Events: Sprint Planning, Daily Scrum, Review, Retrospective.

Example: A food delivery app releases search, payment, and tracking over three sprints instead of waiting for the complete application.

4. DevOps & CI/CD

DevOps bridges development and operations using automation.

CI automatically builds and tests every code commit. CD automates deployment.

Pipeline Example: Git → Build → Unit Test → Integration Test → Deploy to Staging → Production.

5. Real-world Scenarios

Scenario 1: A customer changes requirements every week. Which methodology is suitable?

Scenario 2: A healthcare application requires complete documentation and regulatory approval. Which SDLC model is appropriate?

Scenario 3: Developers commit broken code. Which CI practice can reduce defects?

6. Interview Questions

1. Explain SDLC with an example.
2. Difference between Agile and Waterfall.
3. Why are retrospectives important?
4. Explain CI/CD with a real project example.
5. What risks occur if testing is skipped?